

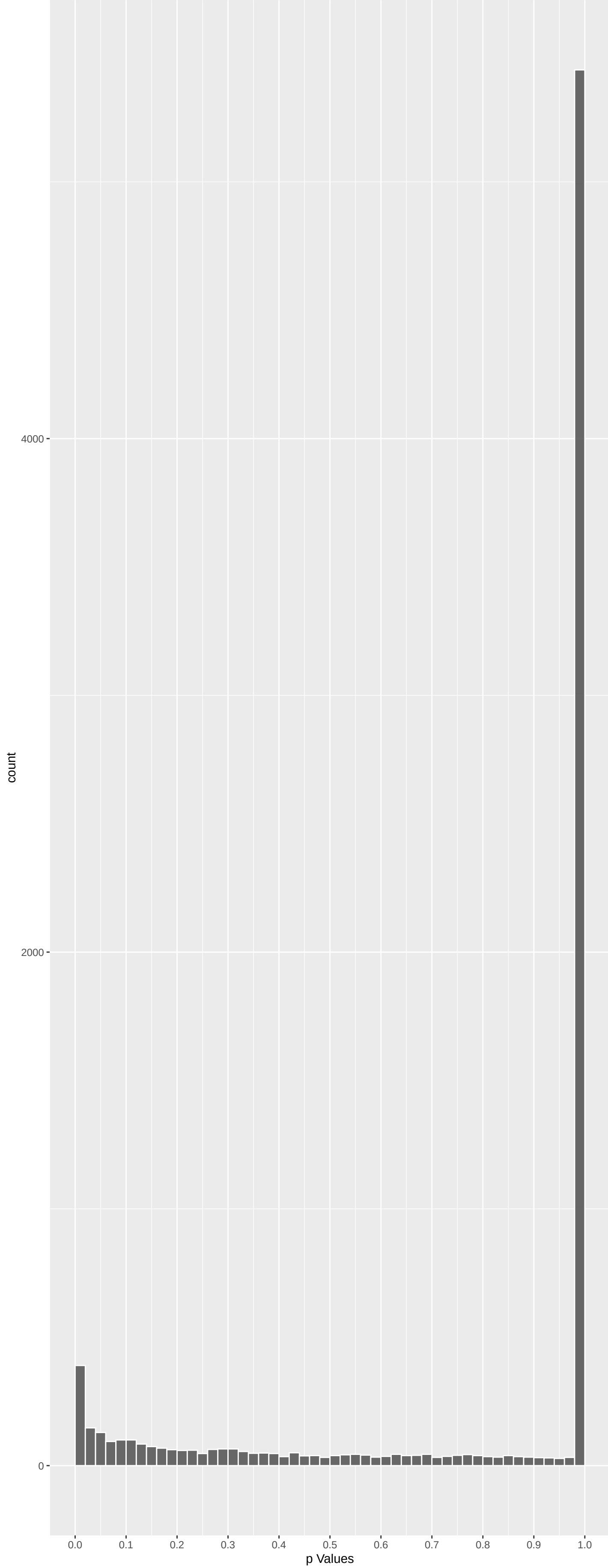
Top genes by P-value non-permulated

Gene	Rho	P	p.adj	qValueNoperm
CLDN16	5.986690	1.284926e-08	1.941e-04	1.000e+00
BPIFB1	5.850289	2.944313e-08	2.224e-04	1.000e+00
CLPS	5.024734	3.024789e-06	7.615e-03	1.000e+00
SLC23A1	5.038034	2.822028e-06	7.615e-03	1.000e+00
CPB1	5.059832	2.517755e-06	7.615e-03	1.000e+00
FAR2	5.065493	2.444072e-06	7.615e-03	1.000e+00
ST6GAL2	4.971277	3.990809e-06	8.612e-03	1.000e+00
EHHADH	4.786481	1.018386e-05	1.725e-02	1.000e+00
HMCES	4.784597	1.027985e-05	1.725e-02	1.000e+00
ALB	4.755973	1.184959e-05	1.790e-02	1.000e+00
GCDH	4.656781	1.927147e-05	1.927e-02	1.000e+00
RBSN	4.645199	2.038493e-05	1.927e-02	1.000e+00
SLC1A1	4.643276	2.057564e-05	1.927e-02	1.000e+00
EPN3	4.632435	2.168343e-05	1.927e-02	1.000e+00
CPAT2	4.698080	1.575711e-05	1.927e-02	1.000e+00
FGD5	4.646397	2.026692e-05	1.927e-02	1.000e+00
SPTA1	4.699564	1.564304e-05	1.927e-02	1.000e+00
PNLIP	4.612707	2.384754e-05	2.001e-02	1.000e+00
IL1R2	4.566210	2.979726e-05	2.369e-02	1.000e+00
COQ9	4.501934	4.040274e-05	3.051e-02	1.000e+00
SLCA4A1	4.469556	4.702936e-05	3.089e-02	1.000e+00
PLPP1	-4.475251	4.579297e-05	3.089e-02	1.000e+00
CHAC2	4.470329	4.685972e-05	3.089e-02	1.000e+00
HIRA	4.408623	6.241801e-05	3.735e-02	1.000e+00
TM4SF4	4.395087	6.643711e-05	3.735e-02	1.000e+00

Top genes by Q-Value non-permulated

Gene	Rho	P	p.adj	qValueNoperm
M6PR	-0.22723602	1.000000000	1.000e+00	1.000e+00
FKBP4	1.48905881	0.818831294	1.000e+00	1.000e+00
CYP26B1	-1.22228561	1.000000000	1.000e+00	1.000e+00
NDUFAF7	-0.94274266	1.000000000	1.000e+00	1.000e+00
FUCA2	-0.01167970	1.000000000	1.000e+00	1.000e+00
DBND1	0.05102369	1.000000000	1.000e+00	1.000e+00
HS3ST1	1.16817078	1.000000000	1.000e+00	1.000e+00
SEMA3F	1.06388121	1.000000000	1.000e+00	1.000e+00
CFTR	-0.98320336	1.000000000	1.000e+00	1.000e+00
CYP51A1	1.87343602	0.366048204	1.000e+00	1.000e+00
USP28	-0.43824354	1.000000000	1.000e+00	1.000e+00
SLC7A2	-0.15814565	1.000000000	1.000e+00	1.000e+00
HSPB6	-0.72199313	1.000000000	1.000e+00	1.000e+00
PK4	3.03067446	0.014640490	5.122e-01	1.000e+00
USH1C	-0.26189262	1.000000000	1.000e+00	1.000e+00
BAIAP2L1	2.35377380	0.1111503495	1.000e+00	1.000e+00
TMEM132A	0.51409698	1.000000000	1.000e+00	1.000e+00
DVL2	-3.14628613	0.009921489	4.295e-01	1.000e+00
RPAP3	-0.83778218	1.000000000	1.000e+00	1.000e+00
HCKA11	-1.70429835	0.529852179	1.000e+00	1.000e+00
TRAPPC6A	-0.11779667	1.000000000	1.000e+00	1.000e+00
SPATA20	-0.68470369	1.000000000	1.000e+00	1.000e+00
CD79B	0.43675758	1.000000000	1.000e+00	1.000e+00
RHBD1	2.60268517	0.055498111	9.608e-01	1.000e+00
RPUSD1	0.83047657	1.000000000	1.000e+00	1.000e+00

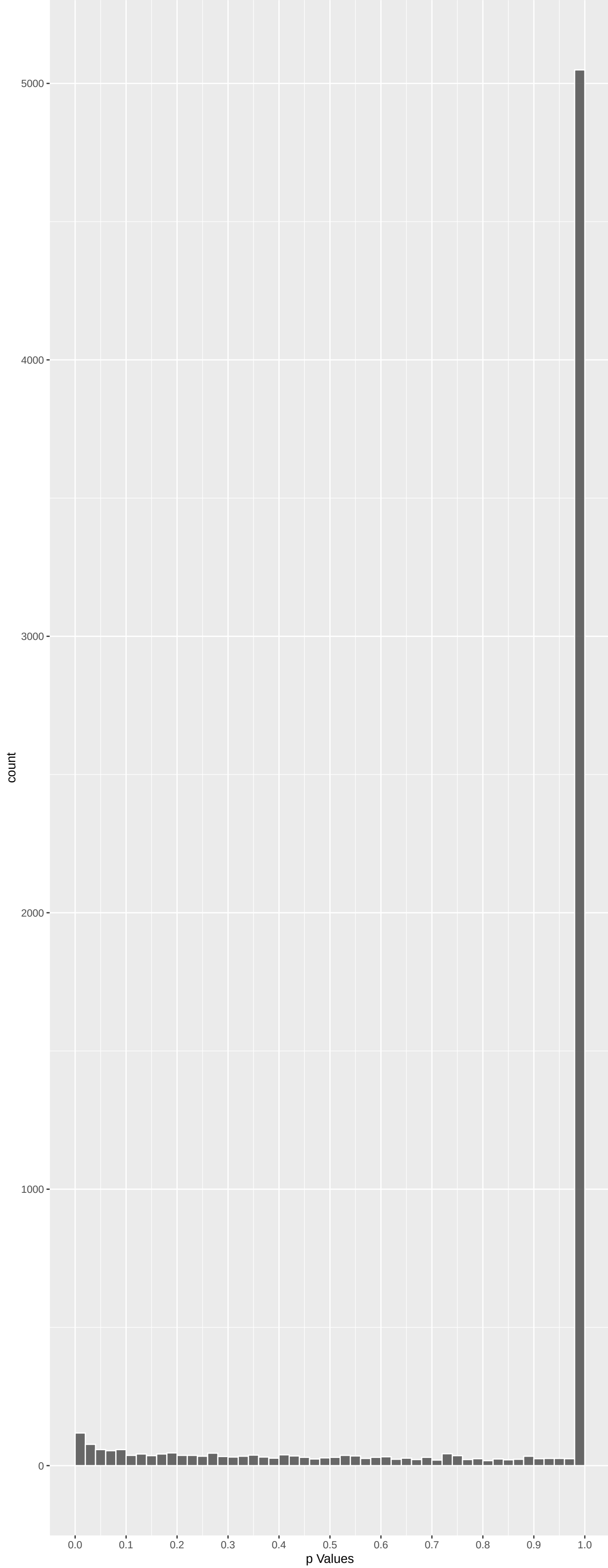
Positive Rho Non-permulated



Top Positive genes by P-value non-permulated

Gene	Rho	P	p.adj	qValueNoperm
CLDN16	5.986690	1.284926e-08	1.941e-04	1.000e+00
BPIFB1	5.850289	2.944313e-08	2.224e-04	1.000e+00
CLPS	5.024734	3.024789e-06	7.615e-03	1.000e+00
SLC23A1	5.038034	2.822028e-06	7.615e-03	1.000e+00
CPB1	5.059832	2.517755e-06	7.615e-03	1.000e+00
FAR2	5.065493	2.444072e-06	7.615e-03	1.000e+00
ST6GAL2	4.971277	3.990809e-06	8.612e-03	1.000e+00
EHHADH	4.786481	1.018386e-05	1.725e-02	1.000e+00
HMCES	4.784597	1.027985e-05	1.725e-02	1.000e+00
ALB	4.755973	1.184959e-05	1.790e-02	1.000e+00

Negative Rho Non-permulated



Top Negative genes by P-value non-permulated

Gene	Rho	P	p.adj	qValueNoperm
PLPP1	-4.475251	4.579297e-05	3.089e-02	1.000e+00
BIRC3	-4.279569	1.123531e-04	4.714e-02	1.000e+00
MORN4	-4.255687	1.250447e-04	5.105e-02	1.000e+00
SNRPD3	-4.129321	2.183020e-04	7.016e-02	1.000e+00
RPL34	-4.062023	2.918952e-04	8.183e-02	1.000e+00
H2BC21	-4.032462	3.311739e-04	8.701e-02	1.000e+00
TRPC4AP	-4.007572	3.680770e-04	8.799e-02	1.000e+00
CFAP20	-3.970383	4.305429e-04	9.105e-02	1.000e+00
HSPB3	-3.970372	4.305629e-04	9.105e-02	1.000e+00
ICMT	-3.946960	4.748980e-04	9.661e-02	1.000e+00

GO_Biological_Process_2023 Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Inorganic Cation Transmembrane Transport	0.08675135	235	5.324e-06	1.614e-02	ANO6.95 SLC6A12:119 SLC36A2:138 HPN1:60 SLC6A15:194 ATP6V0A4:196 ATP2A1:216
Monoatomic Cation Transmembrane Transport	0.08536139	234	7.845e-06	1.614e-02	ANO6.95 SLC6A12:119 SLC36A2:138 HPN1:60 SLC6A15:194 ATP6V0A4:196 ATP2A1:216
Monocarboxylic Acid Transport (GO:001571)	0.18171165	50	9.015e-06	1.614e-02	SLC5A8:64 SLC5A12:67 SLC6A12:119 SLC2A1:67 FABP1:179 KDM5A:41 SLC518:455
Organic Hydroxy Compound Transport (GO:0)	0.20888495	36	1.468e-05	1.971e-02	SLC22A1:62 SLC5A8:64 ABCD3:167 GHR1:252 SLC518:455 SLC51A:698 SLC10A2:869
Carboxylic Acid Transport (GO:0046942)	0.16184370	47	1.259e-04	1.353e-01	SLC5A12:67 SLC6A12:119 SLC5A12:67 SLC5A8:64 SLC5A12:67 SLC6A18:429 SLC6A6:478 SLC4A1:553
Intracellular Monoatomic Ion Homeostasis	0.23354165	19	4.269e-04	3.822e-01	CLDN16:103 CNNM4:204 SLC4A1:312 LETM1:510 SLC39A10:812 RHCG:813 RHAG:855
Organic Anion Transport (GO:0015711)	0.11847502	72	5.213e-04	4.000e-01	SLC05A1:111 SLC52A3:143 SLC4A9:176 SLC4A1:312 SLC4A3:390 MGST1:402 SLC22A13:514
Sodium Ion Transmembrane Transport (GO:0)	0.11890770	67	7.794e-04	5.232e-01	SLC6A12:119 SLC6A15:194 SLC9A4:319 SLC12A3:356 SCNN1B:358 SCNN1B:406 SLC6A18:429
Organic Cation Transport (GO:0015695)	0.17407857	30	9.747e-04	5.817e-01	SLC22A1:62 SLC6A12:119 SLC81B1:126 SLC22A4:675 SLC5A7:766 SLC44A5:769 SLC6A3:845
Bile Acid And Bile Salt Transport (GO:0072041)	0.23220741	16	1.305e-03	5.843e-01	ABCD3:167 SLC518:455 SLC51A:698 SLC10A2:869 SLC10A3:891 SLC27A5:1144 SLC10B1:1202
Nitrogen Compound Transport (GO:0017105)	0.08104306	133	1.301e-03	5.843e-01	SLC22A1:62 SLC5A8:64 SLC6A12:119 SLC36A2:138 SLC52A3:143 SLC6A15:194 CUBN:226
Sulfur Compound Transport (GO:0072348)	0.20871940	20	1.238e-03	5.843e-01	SLC22A1:62 SLC26A4:296 SLC26A8:768 SLC13A1:798 SLC35B3:826 SLC10A2:848 SLC5A6:1422
L-phenylalanine Catabolic Process (GO:00)	0.42004141	4	3.621e-03	6.145e-01	TAT:546 GSTZ1:673 HGD:1024 IL4I1:1697 NA NA NA
L-phenylalanine Metabolic Process (GO:00)	0.42004141	4	3.621e-03	6.145e-01	TAT:546 GSTZ1:673 HGD:1024 IL4I1:1697 NA NA NA
Acid Secretion (GO:0046717)	0.31814180	7	3.561e-03	6.145e-01	DRD3:243 GHR1:252 SLC518:455 SLC13A1:698 SLC26A6:3076 SLC9A3R1:3295 SLC22A16:8031.5
Alpha-Amino Acid Biosynthetic Process (G	0.27832458	10	2.310e-03	6.145e-01	BCAT:28 SDSL:134 CAD:730 SDS:847 CPS1:1137 BCAT:281:2925 OTC:2985
Canonical Wnt Signaling Pathway (GO:0060)	-0.15408568	33	2.206e-03	6.145e-01	PEN5:5 WNT4:25 DVL2:140 FRAT1:358 FZD6:407 BCL9L:411 CSNK1E:4699.5
Erythrose 4-Phosphate/Phosphoenolpyruvat	0.42004141	4	3.621e-03	6.145e-01	TAT:546 GSTZ1:673 HGD:1024 IL4I1:1697 NA NA NA
Ethanol Metabolic Process (GO:0006067)	0.34505051	6	3.425e-03	6.145e-01	ACSS2:412 ADH4:1182 SULT1A2:1487 SULT1E1:2135 SULT1A3:2950 SULT1B1:3463 NA
Gamma-Aminobutyric Acid Transport (GO:00)	0.38776891	5	2.676e-03	6.145e-01	SLC6A12:119 SLC6A6:478 SLC6A11:646 SLC6A13:2492 SLC9A3R1:3295 NA NA
Innate Immune Response In Mucosa (GO:000)	0.42198546	4	3.468e-03	6.145e-01	BP1FB1:2 NOS2:197 PLA2G1B:225 IL4:3480 NA NA NA
Organic Cation Import Across Plasma Me	0.09640985	85	2.127e-03	6.145e-01	SLC12A9:271 CLU:299 SLC12A6:305 SLC9A4:319 SLC12A3:356 SCNN1G:358 SCNN1B:406
Monoatomic Cation Homeostasis (GO:005058)	0.11232214	61	2.450e-03	6.145e-01	CLDN16:103 CNNM4:204 SLC12A9:271 SLC12A6:305 SLC12A3:356 SCNN1G:358 SCNN1B:406
Mucosal Immune Response (GO:002385)	0.28314470	10	1.935e-03	6.145e-01	BP1FB1:2 NOS2:197 PLA2G1B:225 FFA2:426 FJNLRI:1338 RAB17:2714 PIGR:2961
Negative Regulation Of Gene Expression V	0.3763612	5	3.562e-03	6.145e-01	ZNF445:338 UHRF1:1311 USP7:1572 HELLS:1669 UHRF2:2788 NA NA
Negative Regulation Of miRNA Metabolic P	0.21677282	15	3.661e-03	6.145e-01	POU5F1:270 PDGFB:603 ESRI1:1493 H1F1A:1560 NFATC4:1662 NFATC3:1690 TGFBI:2288
Negative Regulation Of mRNA Translation P	0.24167076	14	1.748e-03	6.145e-01	POU5F1:270 PDGFB:603 ESRI1:1493 H1F1A:1560 NFATC4:1662 NFATC3:1690 TGFBI:2288
Negative Regulation Of Single Stranded V	0.34856061	6	3.111e-03	6.145e-01	TASOR:5 MORC2:30 INPP5K:534 SETDB1:1161 MPHOSPH8:1841 HMGGA2:8031.5 NA
Positive Regulation Of DNA Methylation-D	0.31902278	8	1.782e-03	6.145e-01	TASOR:5 MORC2:30 SETDB1:1161 TRIM28:1724 MPHOSPH8:1841 PPHLNI:2236 SETDB2:3345
Positive Regulation Of Heterochromatin F	0.26843169	10	3.294e-03	6.145e-01	TASOR:5 MORC2:30 SETDB1:1161 TRIM28:1724 MPHOSPH8:1841 PPHLNI:2236 SETDB2:3345
Regulation Of Cell Adhesion (GO:0030155)	-0.08285646	105	3.441e-03	6.145e-01	TGM2:44 PAH8:70 RAC2:161 CXCR3:394 FUT2:400 FES:402 NA NA NA
Sodium Ion Transport (GO:0006814)	0.10041948	74	2.872e-03	6.145e-01	SLC5A8:64 SLC5A12:67 SLC6A12:119 SLC6A15:194 SLC12A3:356 SCNN1G:358 SCNN1B:406
One-Carbon Compound Transport (GO:001975)	0.13657119	29	4.205e-03	6.643e-01	AQP11:35 SLC4A9:176 SLC4A1:312 SLC9A3:390 RHAG:855 CAD:747 CA4:1290
Sulfate Transport (GO:0008272)	0.2602506	10	4.121e-03	6.643e-01	SLC26A4:296 SLC26A8:768 SLC13A1:798 SLC25A10:848 SLC26A2:2397 SLC13A4:2561 SLC12A6A:3075
Chromosome Condensation (GO:0030261)	0.18648092	19	4.909e-03	7.202e-01	H4:6-139 ACFM:581:527 PLK1:811 AFM1:863 ERN2:964 NCAPH:1141 TOP2A:1830
Fat-Soluble Vitamin Metabolic Process (G	0.20373483	16	4.793e-03	7.202e-01	CYP11A1:4 LRAT:112 NQO1:374 KDM5A:401 GGX5:505 CYP1A1:695 P4SA4:1135
Intracellular pH Elevation (GO:0051454)	0.30573074	7	5.095e-03	7.202e-01	NOX1:213 TMEM175:567 CLN3:1383 SLC6A3:1580 COA2:2165 SLC26A6:3075 SLC45A2:8031.5
Replication Fork Processing (GO:0031297)	-0.13348788	37	4.995e-03	7.202e-01	BRCA1:247 RAD51:284 MUS81:369 PARP1:4699.5 ATRX:4699.5 WRN:4699.5 FBH1:4699.5
Aromatic Amino Acid Family Catabolic Pro	0.26303706	9	6.292e-03	8.242e-01	TAT:546 GSTZ1:673 HGD:1024 IL4I1:1697 HGO:1024 IL4I1:1697 NA NA NA
Sodium Ion Homeostasis (GO:0055078)	0.19151404	17	6.278e-03	8.242e-01	SLC12A3:356 SCNN1G:358 SCNN1B:406 ATP1B3:477 SCNN1A:653 ATP4B:1118 TMPRSS4:1179

EnrichmentHsSymbolsFile2 Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
REACTOME_SLC_MEDIATED_TRANSMEMBRANE_TRAN	0.12279528	219	4.404e-10	2.852e-06	SLC1A1:12 SLC22A1:62 SLC5A8:64 SLC5A12:67 SLC6A12:119 SLC36A2:138 SLC2A9:173
REACTOME_TRANSPORT_OF_SMALL_MOLECULES	0.06514096	96	1.566e-07	3.382e-04	SLC1A1:12 AQP11:35 MTPP:36 ALB:51 SLC22A1:62 SLC5A8:64 SLC5A12:67
WP_NRF2_PATHWAY	0.16059022	52	1.077e-07	3.382e-04	TXNRD3:58 SLC5A8:64 SLC5A12:67 MGST3:80 SLC2A9:173 SLC6A15:194 TXNRD1:282
REACTOME_TRANSPORT_OF_INORGANIC_CATIONS_	0.14702944	95	7.612e-07	1.232e-03	SLC1A1:12 SLC5A8:64 SLC5A12:67 SLC6A12:119 SLC36A2:138 SLC2A9:173 SLC6A15:194
WP_NUCLEAR_RECEPTORS_METAPATHWAY	0.09475113	222	1.257e-06	1.628e-03	EHHADH:29 ABCB1:56 TXNRD3:58 SLC5A8:64 SLC5A12:67 MGST3:80 CYP7A1:131
REACTOME_TRANSPORT_OF_BILE_SALTS_AND_ORG	0.15728318	77	1.880e-06	1.951e-03	SLC22A1:62 SLC6A12:119 SLC6A15:194 SLC13A2:205 SLC6A18:429 SLC6A6:478 SLC39A2:566
REACTOME_SLC_TRANSPORTER_DISORDERS	0.14901738	85	2.109e-06	1.951e-03	SLC1A1:12 SLC36A2:138 NUP133:147 SLC2A9:173 SLC26A4:296 SLC12A6:305 SLC4A1:553
CAIRO_HEPATOBLASTOMA_DT	0.09447100	196	5.490e-06	4.444e-03	SLC1A1:12 GYS2:34 ABCB1:56 SLC22A1:62 GCDH:78 HP:90 ILIRAP:97
WP_TRANSULFURATION_ONECARBON_METABOLISM	0.18126636	51	5.745e-06	5.501e-03	BCAT:28 CHDH:127 GNMT:174 PHGDH:301 GPX4:371 TYMS:468 SARDH:634
FEVR_CTNNB1_TARGETS_UP	0.05261648	502	6.327e-05	3.854e-02	CLPS1:1 SLC1A1:12 ENTPD5:20 SPTA1:33 AQP11:35 MTPP:36 PRSS4:39
LEE_LIVER_CANCER_DENA_DN	0.14481559	63	7.142e-05	3.854e-02	EHHADH:29 PNLP:37 GCDH:78 ILIRAP:97 CYP7A1:131 ABCD3:167 FABP1:179
HSIAO_LIVER_SPECIFIC_GENES	0.08571511	183	6.695e-05	3.854e-02	GYS2:34 MTPP:36 ALB:51 LGAL5:60 TM4SF:68 HP:90 ATP13A3:113
REACTOME_BIOLOGICAL_OXIDATIONS	0.10142111	123	1.058e-04	4.952e-02	CYP11A1:4 PODXL:266 MGST3:80 CYP7A1:131 COMT:180 CYP2F1:253 MGST1:402
REACTOME_METABOLISM_OF_LIPIDS	0.04752273	581	1.071e-04	4.952e-02	CYP11A1:4 FAR2:18 PIP5K1A:25 EHHADH:29 MORC2:30 ALB:51 ABCB1:56
REACTOME_SIGNALING_BY_WNT	-0.08640085	161	1.614e-04	6.967e-02	WNT4:25 DVL2:140 RAC2:161 H3C4:239 VPS26A:283 FRAT1:358 FZD6:407
REACTOME_AMINO_ACID_TRANSPORT_ACROSS_THE	0.19410420	30	2.348e-04	9.504e-02	SLC6A12:119 SLC36A2:138 SLC6A15:194 SLC6A18:429 SLC6A6:478 SLC43A1:553 SLC3A1:861
WP_Q13_COPY_NUMBER_VARIATION_SYNDROME	0.16072059	43	2.680e-04	1.021e-01	MERTK:70 ILIRL2:87 ILIRAP:97 ZC3H6:245 TTL:373 PSD4:380 IL1A:531
KEGG_STEROID_HORMONE_BIOSYNTHESIS	0.20544983	26	2.889e-04	1.040e-01	CYP11A1:4 CYP7A1:131 COMT:180 HSD17B3:574 HSD17B8:679 SULT2B1:692 CYP1A1:695
WP_OXIDATIVE_STRESS_RESPONSE	0.20027966	27	3.170e-04	1.081e-01	NOX1:213 TXNRD1:282 NOX1:374 MGST1:402 CYP1A1:695 HMOX1:923 SOD1:1055
WP_METAPATHWAY_BIOTRANSFORMATION_PHASE_I	0.10684806	92	4.049e-04	1.311e-01	CYP11A1:4 MGST3:80 CYP7A1:131 COMT:180 CYP2F1:253 GPX4:371 MGPST1:402
HOSHIDA_LIVER_CANCER_SUBCLASS_S3	0.06908213	210	5.853e-04	1.805e-01	SLC23A1:11 EHHADH:29 SLC6A12:119 MMUT:175 AGXT:719 VD4:264 PLN2:326
REACTOME_DIGESTION_OF_DIETARY_LIPID	0.48407682	4	7.991e-04	2.070e-01	CLPS1:1 PNLP:37 PNLP1PR:136 PNLP2PR:442 NA NA NA
BLALOCK_ALZHEIMERS_DISEASE_DN	-0.03617836	765	7.875e-04	2.070e-01	PTEN:5 ATP6V1C1:10 STXBP1:64 RALB:67 PCDH1X:69 NUDF8:88 SYT11:114
CHIANG_LIVER_CANCER_SUBCLASS_PROLIFERATI	0.08421263	136	7.164e-04	2.070e-01	EHHADH:29 GYS2:34 MTPP:36 ABCB1:56 SLC22A1:62 GCDH:78 DHTKD1:130
WOO_LIVER_CANCER_RECURRENCE_DN	0.12599558	60	7.449e-04	2.070e-01	MTPP:36 GCDH:78 SLC2A9:173 AASS:288 ACADSB:561 SARDH:634 MAT1A:724
REACTOME_DISORDERS_OF_TRANSMEMBRANE_TRAN	0.08259665	133	1.031e-03	2.568e-01	SLC1A1:12 ABCB1:56 SLC6A2:138 NUP133:147 SLC2A9:173 SLC26A4:296 SLC12A6:305
STARK_WHOIPCAMPS_22Q11_DELETION_DN	0.22960447	17	1.079e-03	2.568e-01	HIRA:61 TRMT2A:169 COMT:180 ARVCF:608 MRLP:403 T3S ZDHCH8:933 UF01:1138
REACTOME_HEME_DEGRADATION	0.29579473	10	1.200e-03	2.658e-01	ALB:51 FABP1:179 HMOX1:923 UGT1A1:925 SLC01B1:242 BLVRB:1531 HMOX2:2267
REACTOME_BETA_CATENIN_INDEPENDENT_WNT_SI	-0.10463927	80	1.231e-03	2.658e-01	WNT4:25 DVL2:140 RAC2:161 FZD6:407 AQP11:35 SLC1A1:12 RAC1:451
REACTOME_SENSORY_PERCEPTION_OF_TASTE	0.17404747	29	1.181e-03	2.658e-01	TAS2R41:71 TAS2R2A:208 SCNN1G:358 TAS1R3:385 SCNN1B:406 TAS1R2:521 SCNN1A:653
WP_FOCAL_ADHESION	-0.07403851	158	1.359e-03	2.825e-01	PTEN:5 BIRC3:58 CRK1:58 RAC2:161 MAPK8:211 VEGFD:282 SHC1:346
BERTUCCI_MEDULLARY_LS_DUCTAL_BREAST_CANC	-0.08427575	121	1.396e-03	2.825e-01	GUCY1A2:127 UACA:170 SNRPB:128 ADAMTSL2:267 SPOCK1:286 ACTN1:4699.5 RGS4:4699.5
ANDERSEN_LIVER_CANCER_KRT19_DN	0.12370233	54	1.678e-03	2.841e-01	GNMT:174 PCYOX1:287 SLENENP1:414 PTGR1:1347 SLC1A1:12 AQP1:779 ALDH1A1:97
CARRILLOREIXACH_HEPATOBLASTOMA_WS_NORMAL	0.03351551	785	1.642e-03	2.841e-01	CYP11A1:4 HMCE5:8 SLC1A1:12 EHHADH:29 GYS2:34 ABCB1:56 SLC22A1:62
KEGG_TASTE_TRANSDUCTION	0.17611416	27	1.543e-03	2.841e-01	TAS2R41:71 TAS2R4:108 SCNN1G:358 TAS1R3:385 SCNN1B:406 ADCY4:460 TAS1R3:521
REACTOME_TRANSLATION	-0.06703756	189	1.535e-03	2.841e-01	RPS8:32 MRLP:38:78 SEC16B:106 MRLP:37:139 FARSA:194 DAP3:213 PRL2:270
REACTOME_DIGESTION_AND_ABSORPTION	0.23391898	15	1.711e-03	2.841e-01	CLPS1:1 PNLP:37 PNLP1PR:136 PNLP2PR:442 GUC2A:558 SLC2A2:899 LCT:1610
YOSHIMURA_FAPK8_TARGETS_UP	0.03201275	877	1.546e-03	2.841e-01	CLPS1:1 SLC1A1:12 GYS2:34 PNLP:37 RBP2:57 LGAL5:60 SLC22A1:62
KEGG_FOCAL_ADHESION	-0.07267361	158	1.666e-03	2.841e-01	PTEN:5 BIRC3:58 CRK1:58 RAC2:161 MAPK8:211 VEGFD:282 SHC1:346
VERNOCHET_AIDOPOGENESIS	0.24823034	13	1.944e-03	2.861e-01	HP:90 APCDD1:157 MC2R:224 CYP2F1:253 FFA2:426 RARRES2:1197 ADCYAP1R1:1601

DisGeNET Top pathways by non-permutation

Left-Sided Heart Failure	0.09029723	94	2.540e-03	7.301e-01	ALB:51 PTH:77 NOS2:197
Liver neoplasms	0.03008995	870	2.531e-03	7.301e-01	SUGP1:217 MT:76 PROM1:40
Nephrolithiasis	0.11583670	76	4.925e-04	7.301e-01	PTH:77 CLDN16:103
Persistent truncus arteriosus	0.20322111	19	1.173e-03	7.301e-01	HIRA:61 GJAS:145
Recurrent infections	0.13487133	48	1.242e-03	7.301e-01	PRKCD:91 MANBA:104
Reticulocytosis	0.21924926	20	6.911e-04	7.301e-01	SEC23B:31 SLC4A1:312
Sea-Blue Histiocyte Syndrome	0.41437531	5	1.333e-03	7.301e-01	NPC1:116
Seborrheic dermatitis	0.22741693	18	8.401e-04	7.301e-01	HIRA:61 COMT:180
Spherocytosis	0.39463988	5	2.243e-03	7.301e-01	SPTA1:33 SLC4A1:312
Taste sweet	0.25991207	12	1.827e-03	7.301e-01	TAS1R1:385
Tetany	0.10330369	19	1.311e-03	7.301e-01	HIRA:61 PTH:77
Adenocarcinoma of rectum	0.17318075	24	3.330e-03	7.550e-01	MYD88:125
Cholangiolocellular Carcinoma	0.14798942	4	3.789e-03	7.550e-01	PROM1:40
Drug-Induced Liver Disease	0.06206994	183	3.947e-03	7.550e-01	IL1R1:27
Epidermolysis Bullosa	-0.15632613	30	3.061e-03	7.550e-01	KLHL24:83
Essential Hypertension	0.05963516	199	3.908e-03	7.550e-01	ALB:51 PTH:77
Hyperactive renin-angiotensin system	0.29450108	8	3.924e-03	7.550e-01	SCNN1G:358

KeggReactome Top pathways by non-permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
REACTOME_OXIDATIVE_STRESS_INDUCED_SENESC	-0.21402315	76	1.231e-10	2.220e-07	H2BC216 RPS27A:28
REACTOME_MEIOTIC_RECOMBINATION	-0.23869030	47	1.574e-08	1.419e-05	H2BC216 H4C2:54
REACTOME_PRC2_METHYLATES_HISTONES_AND_N	-0.26602096	36	3.435e-08	1.690e-05	H2BC216 H4C2:54
REACTOME_SLC_MEDATED_TRANSMEMBRANE_DTR	0.10491174	232	4.686e-08	1.690e-05	SLC1A1:16
REACTOME_DNA_METHYLATION	-0.29999799	28	4.017e-08	1.690e-05	H2BC216 H4C2:54
REACTOME_ERCC8_CSB_AND_EHMT2_GBA_POSITIV	-0.25413035	37	9.119e-08	2.740e-05	H2BC216 H4C2:54
REACTOME_SIRT1_NEGATIVELY_REGULATES_RRNA	-0.23796682	1	1.369e-07	3.525e-05	H2BC216 H4C2:54
REACTOME_TRANSPORT_OF_SMALL_MOLECULES	0.06286240	623	1.589e-07	3.581e-05	ALB:10
REACTOME_TRANSPORT_OF_INORGANIC_CATIONS	0.14952996	101	2.268e-07	4.544e-05	SLC1A1:16
REACTOME_ASSEMBLY_OF_THE_ORC_COMPLEX_AT	-0.27531571	29	2.939e-07	5.299e-05	H2BC216 H4C2:54
REACTOME_HBD5_DEACETYLATE_HISTONES	-0.20485244	50	5.630e-07	8.459e-05	H2BC216 H4C2:54
KEGG_VALINE_LEUCINE_AND_ISOLEUCINE_DEGRA	0.22137781	43	5.267e-07	8.459e-05	EHADH:48
REACTOME_CELLULAR_SENESCENCE	-0.12287383	135	9.121e-07	1.107e-04	H2BC216 RPS27A:28
REACTOME_POSITIVE_EPIGENETIC_REGULATION	-0.18679392	58	9.031e-07	1.107e-04	H2BC216 H4C2:54
REACTOME_TRANSCRIPTIONAL_REGULATION_BY_R	-0.11593708	152	9.160e-07	1.107e-04	H2BC216 RPS27A:28
REACTOME_BIOLOGICAL_OXIDATIONS	0.11686801	145	1.332e-06	1.417e-04	CHAC2:21
REACTOME_RNA_POLYMERASE_II_TRANSCRIPTION	-0.04988198	867	1.336e-06	1.417e-04	SNRPD3
REACTOME_METABOLISM_OF_LIPIDS	0.05648986	650	1.557e-06	1.560e-04	FAR2:3
REACTOME_INTRACELLULAR_SIGNALING_BY_SCO	-0.08809219	250	1.968e-06	1.867e-04	MAPKAP1:31
REACTOME_MEIOSIS	-0.15776422	75	2.444e-06	2.203e-04	H2BC216 H4C2:54
REACTOME_ACTIVATED_PKNI_STIMULATES_TRANS	-0.26116916	27	2.683e-06	2.304e-04	H2BC216 H4C2:54
REACTOME_RNA_POLYMERASE_II_PROMOTER_ESCAP	-0.20054661	45	3.347e-06	2.624e-04	H2BC216 H4C2:54